USER MANUAL

Invoice Management Application (Invoice App)

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1. Introduction

The **Invoice App** is an invoice creation and management system developed using the Python programming language. The application features a user-friendly Graphical User Interface (GUI) built with the Tkinter library and uses an SQLite database to store customer, invoice, and product information. In addition, it provides the ability to export invoices in PDF format.

The primary purpose of the application is to simplify the process of creating and managing invoices by providing an intuitive interface for customer registration, product entry, and automatic calculation of financial values.

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2. System Requirements

To ensure the proper operation of the application, the following requirements must be met:

- Python 3.10 or later
- Windows, Linux, or macOS operating system
- The following Python libraries:
    - Tkinter
    - SQLite3
    - ReportLab
    - tkcalendar

The required external libraries can be installed using the following commands:

```
pip install reportlab
pip install tkcalendar
```

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3. Starting the Application

The application is launched by running the **main.py** file.

During its first execution, the application automatically creates the **invoice.db** database, where all customer, invoice, and product data are stored.

After the application starts, the main application window is displayed.

(Figure 1 – Main Application Window will be inserted here.)

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4. Main Window Description

The main window consists of the following sections.

Customer Selection

At the top of the window, a drop-down list displays all registered customers. The user selects the customer for whom the invoice will be created.

Next to the customer list is the **Add Client** button, which allows the user to register a new customer.

A date selection field is also available for choosing the invoice date.

Product Table

The center of the application contains a table where the products or services included in the invoice are entered.

Each row displays the following information:

- Description
- Quantity
- Price
- Total Cost

Product Entry Section

To add a product or service, the user fills in the following fields:

- Description
- Qty
- Price

After entering the information, the user clicks the **Add** button.

The product is automatically inserted into the invoice table.

The **Remove** button deletes the selected product(s) from the invoice.

Calculation Section

At the bottom of the application, the following values are displayed:

- Subtotal
- VAT (%)
- Total

These values are calculated automatically whenever products are added or removed.

Invoice List

The lower section of the main window displays all saved invoices.

Each invoice contains the following information:

- Invoice ID
- Date
- Customer
- Total Amount

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5. Creating a New Customer

To register a new customer, follow these steps:

1. Click the **Add Client** button.
2. Fill in the following information:
    - First Name
    - Last Name
    - Email
    - Phone Number
    - Address
    - Postal Code
    - Tax Identification Number (TIN)
3. Click the **Save** button.

If another customer with the same Tax Identification Number (TIN) already exists, the application displays an error message and the customer record is not saved.

(Figure 2 – Add Client Window will be inserted here.)

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6. Creating a New Invoice

To create a new invoice:

1. Select a customer.
2. Choose the invoice date.
3. Enter the description of the product or service.
4. Enter the quantity.
5. Enter the unit price.
6. Click the **Add** button.

These steps can be repeated until all products or services have been added.

The application automatically updates the subtotal, VAT amount, and final total.

(Figure 3 – Adding Products to an Invoice will be inserted here.)

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7. Saving an Invoice

After all products have been added, click the **Save Invoice** button.

The application performs the following actions:

- Saves the invoice in the database.
- Stores all invoice items.
- Displays a confirmation message indicating that the invoice has been saved successfully.
- Automatically refreshes the invoice list.

(Figure 4 – Invoice Saved Confirmation will be inserted here.)

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8. Viewing Saved Invoices

All saved invoices are displayed in the invoice list.

By double-clicking an invoice, a new window opens displaying:

- all products,
- quantities,
- prices,
- subtotal,
- VAT amount,
- total amount.

This feature allows the user to review the invoice before exporting it.

(Figure 5 – Invoice Preview will be inserted here.)

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9. Exporting an Invoice to PDF

The application allows any saved invoice to be exported as a PDF document.

The export procedure is as follows:

1. Select an invoice from the invoice list.
2. Click the **Export PDF** button.
3. Choose the destination folder.
4. Enter the desired file name.

The generated PDF document contains:

- Invoice number
- Invoice date
- Customer information
- Tax Identification Number (TIN)
- Product table
- Subtotal
- VAT amount
- Grand Total

(Figure 6 – Exporting an Invoice to PDF will be inserted here.)

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10. Database

The application uses an SQLite database for data storage.

The database consists of the following tables:

Clients

Stores:

- Customer information
- Contact details
- Tax Identification Number (TIN)

Invoices

Stores:

- Customer ID
- Invoice date
- Subtotal
- VAT amount
- Total amount

Items

Stores:

- Product or service description
- Quantity

- Unit price
- Total cost

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11. Troubleshooting

No Customer Selected

If no customer has been selected, the application displays an error message requesting the user to select a customer before saving the invoice.

Invalid Quantity or Price

If the quantity or price entered is not a valid numeric value, the application displays an appropriate error message.

Duplicate Tax Identification Number

If a customer is entered using a Tax Identification Number that already exists in the database, the application prevents the record from being saved and displays an error message.

No Invoice Selected

To export an invoice as a PDF, the user must first select a saved invoice from the invoice list.

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12. Conclusion

The **Invoice App** is a comprehensive invoice management application designed for small businesses, freelancers, and educational purposes. It provides an intuitive graphical user interface, secure data storage using SQLite, automatic calculation of financial information, and the ability to generate professional PDF invoices.

The application is easy to use and does not require advanced technical knowledge, making it suitable for a wide range of users. Its straightforward workflow and automated calculations help improve efficiency while reducing the possibility of human error during invoice creation and management.